A degree-sum condition for paths through prescribed vertices*

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Abstract

Let G be a graph of order n and let $S \subseteq V(G)$. Denote by $\kappa(S)$ the minimum size of a vertex set separating two vertices of S, and by $\sigma_p(S)$ the minimum degree sum over all independent sets of p vertices in S. We prove that if $\kappa(S) \geq k \geq 2$ and $\sigma_p(S) \geq n + (p-1)^2$ for every integer $p \geq k$, then any two distinct vertices of S are joined by a path containing all vertices of S. This gives a path version of Ota's theorem on cycles containing a prescribed set of vertices. As an application, we prove a conjecture of Faudree, Faudree, Gould, Jacobson and Magnant: for $k \geq 3$, every k-connected graph G of order n satisfying $\delta(G) > (n + k^2 - 2k)/k$ and $\delta(G) \geq \alpha(G) + k - 2$ is Hamiltonian-connected.

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1 Introduction

We consider finite simple graphs and use standard terminology and notation from [1] and [18]. For a graph $G = (V(G), E(G))$, let $\delta(G)$, $\kappa(G)$, and $\alpha(G)$ denote its minimum degree, connectivity, and independence number, respectively. For $x \in V(G)$, write $N_G(x)$ for the neighborhood of x and $d_G(x) = |N_G(x)|$ for its degree. For two distinct vertices x and y in G, an (x, y)-*path* is a path whose endpoints are x and y. A Hamilton cycle (resp., path) in G is a cycle (resp., path) containing every vertex of G. A graph is *hamiltonian* if it contains a Hamilton cycle, and it is *hamiltonian-connected* if every two distinct vertices are joined by a Hamilton path.

The existence of Hamilton cycles is a classical topic in graph theory. Dirac [5] proved that every graph G of order $n \geq 3$ with $\delta(G) \geq n/2$ is hamiltonian. Ore [15] replaced Dirac's condition by $d_G(x) + d_G(y) \geq n$ for every pair of nonadjacent vertices x and y. A well-known result of Chvátal and Erdős [4] from 1972 states that a graph whose connectivity is at least its independence number is hamiltonian, and that a graph whose connectivity is at least one more than its independence number is hamiltonian-connected.

Thus degree, connectivity, and independence number provide different ways to ensure a Hamilton cycle or a Hamilton path. See [13] for a survey of path and cycle results based on connectivity and independence number. For a k-connected graph, the Chvátal–Erdős theorem [4] ensures Hamiltonicity when $\alpha(G) \leq k$. Under an additional minimum-degree condition, Faudree, Faudree, Gould, Jacobson and Magnant [8, Proposition 1] replaced this bound by $\alpha(G) \leq \delta(G) - k + 2$. They proved that: If G is a k-connected graph of order n with $\delta(G) > (n + k^2 - k - 1)/(k + 1)$ and $\delta(G) \geq \alpha(G) + k - 2$, where $k \geq 2$, then G is hamiltonian. To study the hamiltonian-connectedness, they proposed the following conjecture.

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Conjecture 1.1 (Faudree–Faudree–Gould–Jacobson–Magnant [8]). *Let G be a k-connected graph of order n, where $k \geq 3$. If $\delta(G) > (n + k^2 - 2k)/k$ and $\delta(G) \geq \alpha(G) + k - 2$, then G is hamiltonian-connected.*

As a first step, they showed that Conjecture 1.1 holds in the following two cases.

Theorem 1.2 (Faudree–Faudree–Gould–Jacobson–Magnant [8]). *Let $k \geq 3$, and let G be a graph of sufficiently large order n satisfying $\delta(G) > (n + k^2 - 2k)/k$ and $\delta(G) \geq \alpha(G) + k - 2$. Then G is hamiltonian-connected if one of the following holds:*

(i) $\kappa(G) \geq 4k^2 + 1$;
(ii) $k \in \{3, 4\}$ *and* $\kappa(G) \geq k$.

Thus the cases $k = 3, 4$ for Conjecture 1.1 were known for sufficiently large graphs, while the general result required $\kappa(G) \geq 4k^2 + 1$. The gap $k \leq \kappa(G) < 4k^2 + 1$ and the restriction on the order were left open.

There is also a prescribed-vertex version of the hamiltonian problem. A set $S \subseteq V(G)$ is called *cyclable* if there exists one cycle of G containing every vertex of S. Dirac's theorem [6] states that every k specified vertices of a k-connected graph lie on a common cycle. It led to many later results on paths and cycles through specified vertices; see, for example, [7, 9–11].

Let G be a graph and let $S \subseteq V(G)$ with $|S| \geq 2$. A set $U \subseteq V(G) \setminus \{x, y\}$ *separates* distinct vertices x, y if x and y lie in different components of $G - U$. If $G[S]$ is not complete, define

$$\kappa(S) = \min\{|U| : U \text{ separates two vertices of } S\}.$$

If $G[S]$ is complete, define $\kappa(S) = |S| - 1$. In particular, $\kappa(V(G)) = \kappa(G)$, where $\kappa(K_n) = n - 1$. Write $\alpha(S)$ for the maximum cardinality of an independent subset of S. For $1 \leq p \leq \alpha(S)$, define

$$\sigma_p(S) = \min\left\{\sum_{x \in X} d_G(x) : X \subseteq S,\ X \text{ is independent},\ |X| = p\right\}.$$

Write $\sigma_p(G) = \sigma_p(V(G))$ for short. In 1995, Ota obtained the following degree-sum condition for cyclability.

Theorem 1.3 (Ota [16]). *Let G be a graph of order n and let $S \subseteq V(G)$ with $\kappa(S) \geq k \geq 2$. If $\sigma_p(S) \geq n + (p - 1)(p - 2)$ for every integer $p \geq k + 1$, then S is cyclable.*

Theorem 1.3 has hamiltonian consequences and led to later degree-sum results; see [2, 17]. Our aim is to prove a path version in which two prescribed vertices are the endpoints. Long paths between prescribed endpoints have also been studied under degree and Chvátal–Erdős type assumptions. Hu, Tian, Wei, Egawa and Hirohata [12] proved a Fan-type lower bound for a path joining any prescribed pair of vertices in a 3-connected graph. Chen, Hu and Wu [3] later proved a related Chvátal–Erdős type theorem. Those results use hypotheses on the whole graph and seek long paths between two arbitrary endpoints. Our main result uses degree sums only for independent subsets of S and requires a path passing through all vertices of S.

Theorem 1.4. *Let G be a graph of order n and let $S \subseteq V(G)$ with $|S| \geq 2$. If $\kappa(S) \geq k \geq 2$ and*

$$\sigma_p(S) \geq n + (p - 1)^2 \text{ for every integer } k \leq p \leq \alpha(G), \tag{1}$$

then for every two vertices $u, v \in S$, there is a (u, v)-path containing all vertices of S.

Substituting $S = V(G)$ in Theorem 1.4 gives the following.

Corollary 1.5. *Let G be a k-connected graph, where $k \geq 2$. If*

$$\sigma_p(G) \geq n + (p - 1)^2 \text{ for every integer } k \leq p \leq \alpha(G), \tag{2}$$

then G is hamiltonian-connected.

Remark 1. The condition in (2), and hence (1), can not be weakened. In fact, if we replace (2) with

$$\sigma_p(G) \geq n + p^2 - 2p \ \text{ for every integer } k \leq p \leq \alpha(G), \tag{3}$$

then let $H = K_k \vee kK_m$. Clearly, $\alpha(H) = \kappa(H) = k$. Thus there exists only one integer p satisfying $k \leq p \leq \alpha(G)$, and $\sigma_k(H) = k\delta(H) = k(k+m-1) = |H| + k^2 - 2k$. Hence H satisfies (3). However, H is not hamiltonian-connected, because its toughness is at most 1.

As an application of Theorem 1.4, we prove Conjecture 1.1. We organize the paper as follows. In Section 2, we present the proof of Theorems 1.4. In Section 3 we prove Conjecture 1.1.

2 Proof of Theorem 1.4

Let G be a graph. For a vertex subset $Y \subseteq V(G)$ and a vertex $x \in V(G)$, let $N_Y(x) = N_G(x) \cap Y$ and let $d_Y(x) = |N_Y(x)|$. Let $P = v_1v_2 \cdots v_\ell$ be a path in a graph G. We set

$$v_i^+ := v_{i+1} \ \text{ and } \ v_j^- := v_{j-1},$$

respectively. If $a < b$, we define two subpaths

$$\overrightarrow{P}[v_a, v_b] := v_a v_{a+1} \cdots v_b \ \text{ and } \ \overleftarrow{P}[v_b, v_a] := v_b v_{b-1} \cdots v_a.$$

The notation $\overrightarrow{P}(v_a, v_b]$, $\overrightarrow{P}[v_a, v_b)$, and $\overrightarrow{P}(v_a, v_b)$ means that the endvertex a, the endvertex b, or both endvertices $\{a, b\}$, respectively, are deleted from $\overrightarrow{P}[v_a, v_b]$. We also allow $\overrightarrow{P}[v_a, v_a]$ to be a trivial path; every open or half-open version with equal endpoints is empty. Let

$$v_a\overrightarrow{P}v_b = \overrightarrow{P}[v_a, v_b] \ \text{ and } \ v_b\overleftarrow{P}v_a = \overleftarrow{P}[v_b, v_a].$$

A family of k internally disjoint (x, Y)-paths whose terminal vertices are distinct is referred to as a *k-fan* from x to Y. We denote by $f(x, Y \setminus \{x\})$ the maximum integer k^* such that there exists a k^*-fan from x to Y.

Lemma 2.1. *Let G be a graph and let $S \subseteq V(G)$ with $\kappa(S) \geq k$. Then for every vertex $z \in S$, $f(z, S \setminus \{z\}) \geq \min\{k, |S| - 1\}$.*

Proof. If S is a clique of G, then for any $z \in S$, z is adjacent to every vertex of $S \setminus \{z\}$. Thus there exists a $(|S| - 1)$-fan from z to $S \setminus \{z\}$, as desired.

Next we assume that $G[S]$ is not a complete graph. Let G' be a graph obtain from G by adding a new vertex w and joining w to every vertex of $S \setminus \{z\}$. Suppose to the contrary that there exists no $\min\{k, |S| - 1\}$-fan in G from z to $S \setminus \{z\}$, for some vertex $z \in S$. Then Menger's theorem [14] implies that there exists a separating set $U \subseteq V(G') \setminus \{z, w\}$ with $|U| \leq \min\{k, |S| - 1\} - 1$ which separates z and w. Hence, in $G - U$, there exists a vertex in $S \setminus (U \cup \{z\})$ that belongs to a component different from the one containing z. Therefore, $\kappa(S) \leq |U| \leq k - 1$, a contradiction. □

Next we prove Theorem 1.4.

Proof of Theorem 1.4. Let G be a graph of order n, and let $S \subseteq V(G)$ with $|S| \geq 2$. Assume that $\kappa(S) \geq k \geq 2$ and $\sigma_p(S) \geq n + p^2 - 2p + 1$ for any integer p with $k \leq p \leq \alpha(G)$.

We use induction on $|S|$. If $|S| = 2$, then by $\kappa(S) \geq k \geq 2$, we are done. Next we assume that $|S| \geq 3$ and the result holds for every vertex subset T with $2 \leq |T| \leq |S| - 1$. Now we are to show that for any two vertices $u', v' \in S$, there exists an (u', v')-path in G containing all vertices of S.

Suppose to the contrary that there are two vertices $u, v \in S$ for which no (u,v)-path in G contains all vertices of S. We choose the vertex $w \in S \setminus \{u,v\}$ such that $f(w, S \setminus \{w\})$ is as large as possible. Note that

$$\kappa(S \setminus \{w\}) \geq \kappa(S) \geq k \text{ and } \sigma_p(S \setminus \{w\}) \geq \sigma_p(S) \geq n + p^2 - 2p + 1 \text{ for any } p \geq k.$$

Then by the induction hypothesis, there exists a (u,v)-path P in G such that $S \setminus \{w\} \subseteq V(P)$. Clearly, $w \notin V(P)$. Based on $\kappa(S) \geq k$ and Lemma 2.1, we have that there exist $q := f(w, S \setminus \{w\}) \geq \min\{k, |S|-1\}$ paths $P_1, \ldots, P_q$ from w to $V(P)$ such that $V(P_i) \cap V(P_j) = \{w\}$ for $i \neq j$. Let a_i be the vertex of $V(P_i) \cap V(P)$, and assume that $a_1, \ldots, a_q$ appear in this order along $\overrightarrow{P}$. We choose $P_1, \ldots, P_q$ such that $\sum_{i=2}^{q} \left|\overrightarrow{P}[a_1, a_i]\right|$ is minimum.

Fact 1. *$V\left(\overrightarrow{P}(a_{i-1}, a_i)\right) \cap S \neq \emptyset$ for any $2 \leq i \leq q$.*

Proof of Fact 1. Suppose to the contrary that there exists an integer i such that $V\left(\overrightarrow{P}(a_{i-1}, a_i)\right) \cap S = \emptyset$ and $2 \leq i \leq q$. Then

$$u\overrightarrow{P}a_{i-1}P_{i-1}wP_ia_i\overrightarrow{P}v$$

is a (u,v)-path in G passing through all vertices of S, a contradiction. This proves Fact 1. □

By Fact 1, for each $2 \leq i \leq q$, let x_i denote the last vertex of $\overrightarrow{P}(a_{i-1}, a_i)$ that belongs to S. We define x_1 as follows: if $u \neq a_1$, let x_1 be the last vertex of $V\left(\overrightarrow{P}[u, a_1)\right) \cap S$; if $u = a_1$, let $x_1 = u$.

Fact 2. (i) *$|S| \geq k+1$, and hence $q \geq k \geq 2$;*
(ii) *$N(x_j) \cap V\left(\overrightarrow{P}[x_i, a_i)\right) = \emptyset$ for any two distinct integers i, j with $2 \leq j \leq q$ and $1 \leq i \leq q$.*

Proof of Fact 2. (i) Suppose to the contrary that $|S| \leq k$. Since $\{x_2, \ldots, x_q\} \cup \{u, v, w\} \subseteq S$, we have $|S| \geq q+2$, which contradicts the fact that $q \geq \min\{k, |S|-1\} = |S|-1$. This implies that $|S| \geq k+1$, and thus $q \geq \min\{k, |S|-1\} = k$.

(ii) Suppose to the contrary that there exist two distinct integers i, j such that $2 \leq j \leq q$, $1 \leq i \leq q$ and $N(x_j) \cap V\left(\overrightarrow{P}[x_i, a_i)\right) \neq \emptyset$. Let $z \in N(x_j) \cap V\left(\overrightarrow{P}[x_i, a_i)\right)$. If $j < i$, then

$$u\overrightarrow{P}x_jz\overleftarrow{P}a_jP_jwP_ia_i\overrightarrow{P}v$$

is a (u,v)-path passing through all vertices of S, a contradiction. Thus we assume that $i < j$. Then

$$u\overrightarrow{P}zx_j\overleftarrow{P}a_iP_iwP_ja_j\overrightarrow{P}v$$

is a (u,v)-path containing all vertices of S, yielding a contradiction. This proves Fact 2. □

Fact 3. *$\{w\} \cup \{x_2, \ldots, x_q\}$ is an independent set of G.*

Proof of Fact 3. By Fact 2, $x_i \not\sim x_j$ for any two distinct $2 \leq i, j \leq q$. Suppose that there exists an integer $i \in \{2, \ldots, q\}$ such that $w \sim x_i$. Then $u\overrightarrow{P}x_iwP_ia_i\overrightarrow{P}v$ is a (u,v)-path passing through all vertices of S, a contradiction. Therefore $w \not\sim x_j$ for any $2 \leq j \leq q$. This proves Fact 3. □

Fact 4. (i) *$N(x_j) \cap (V(P_i) \setminus \{a_i\}) = \emptyset$ for any two distinct integers i, j with $1 \leq i \leq q$ and $2 \leq j \leq q$;*
(ii) *$N(x_i) \cap (V(P_i) \setminus \{a_i\}) = \emptyset$ for any $2 \leq i \leq q$.*

Proof of Fact 4. (i) Suppose to the contrary that there exist two distinct integers i, j such that $N(x_j) \cap (V(P_i) \setminus \{a_i\}) \neq \emptyset$. Let $z \in N(x_j) \cap (V(P_i) \setminus \{a_i\})$. If $j < i$, then

$$u\overrightarrow{P}x_jzP_iwP_ja_j\overrightarrow{P}v$$

is a (u, v)-path passing through all vertices of S, a contradiction. Next we assume that $i < j$. Then

$$u\overrightarrow{P}x_jzP_iwP_ja_j\overrightarrow{P}v$$

is a (u, v)-path containing all vertices of S, yielding a contradiction. Thus (i) holds.

(ii) Suppose to the contrary that there exists an integer $i \in \{2, \ldots, q\}$ such that $N(x_i) \cap (V(P_i) \setminus \{a_i\}) \neq \emptyset$. Let $z \in N(x_i) \cap (V(P_i) \setminus \{a_i\})$. Then the path wP_izx_i contradicts the choice of $P_1, \ldots, P_q$. This proves Fact 4. □

We write $A := \{a_1, \ldots, a_q\}$, $B := V(P_1) \setminus \{a_1\}$ and $R := V(G) \setminus (V(P) \cup V(P_1))$.

Fact 5. (i) $d_B(w) + \sum_{j=2}^{q} d_B(x_j) \leq |B| - 1$;
(ii) $d_{V(P)}(w) \leq q$.

Proof of Fact 5. (i) From Fact 4(i), we have $N(x_j) \cap (V(P_1) \setminus \{a_1\}) = \emptyset$ for each $2 \leq j \leq q$. Therefore, $\sum_{j=2}^{q} d_B(x_j) = 0$. This, together with the fact that $N_B(w) \subseteq B \setminus \{w\}$, implies $d_B(w) + \sum_{j=2}^{q} d_B(x_j) \leq |B| - 1$, as desired.

(ii) By the choice of w and $P_1, \ldots, P_q$, we have $N(w) \cap V(P) \subseteq \{a_1, \ldots, a_q\}$ and hence $d_{V(P)}(w) \leq q$. This proves Fact 5. □

Fact 6. (i) $N(w) \cap N(x_i) \cap R = \emptyset$ *for any integer* $i \in \{2, \ldots, q\}$;
(ii) $N(x_j) \cap N(x_i) \cap R = \emptyset$ *for any two distinct* $i, j \in \{2, \ldots, q\}$;
(iii) $d_R(w) + \sum_{i=2}^{q} d_R(x_i) \leq |R|$.

Proof of Fact 6. (i) Suppose to the contrary that there exists a vertex $z \in N(x_i) \cap R$ and $z \sim w$, where $i \in \{2, \ldots, q\}$. Then by Fact 4(i)-(ii), we have that $z \notin \bigcup_{i=1}^{q} V(P_i)$. Hence,

$$u\overrightarrow{P}x_izwP_ia_i\overrightarrow{P}v$$

is a (u, v)-path passing through all vertices of S, a contradiction.

(ii) Suppose to the contrary that there exists a vertex $z \in N(x_i) \cap R$ and $z \sim x_j$, where $i, j \in \{2, \ldots, q\}$ and $i \neq j$. Without loss of generality, we assume that $j < i$. Then by Fact 4(i)-(ii), we have that

$$u\overrightarrow{P}x_jzx_i\overleftarrow{P}a_jP_jwP_ia_i\overrightarrow{P}v$$

is a (u, v)-path passing through all vertices of S, a contradiction.

(iii) From (i) and (ii), we have that

$$d_R(w) + \sum_{i=2}^{q} d_R(x_i) = \left|(N(w) \cap R) \cup \left(\bigcup_{i=2}^{q}(N(x_i) \cap R)\right)\right| \leq |R|.$$

This completes the proof of Fact 6. □

A subpath $P^* = \overrightarrow{P}[z, z')$ of P is called *a good subpath* if it satisfies the following:

(1) $z \in S \cup A$ and $z' \in S \setminus A$;
(2) $V\left(\overrightarrow{P}(z, z')\right) \cap (S \setminus A) = \emptyset$.

Write

$$\mathcal{P} = \{v\} \cup \{P^* : P^* \text{ is a good subpath of } P\}.$$

Hence,

$$|\mathcal{P}| \geq q-1 \text{ and } V(P) = \bigcup_{i=1}^{q} V\left(\overrightarrow{P}[x_i, a_i)\right) \cup \bigcup_{P^* \in \mathcal{P}} V(P^*).$$

Fact 7. *Let $P^* \in \mathcal{P} \setminus \{v\}$, and assume that $P^* = \overrightarrow{P}[z, z') \subseteq \overrightarrow{P}[a_s, a_{s+1}]$, where $1 \leq s \leq q-1$. Let $g \in V(P^*)$. The following properties hold:*

(i) *If $g \sim x_i$ for some $i \geq s+1$, then for any $j \geq i+1$, $N(x_j) \cap V\left(\overrightarrow{P}(g, z')\right) = \emptyset$;*

(ii) *If $g \sim x_i$ for some $i \leq s$, then for any $i+1 \leq j \leq s$, $N(x_j) \cap V\left(\overrightarrow{P}(g, z')\right) = \emptyset$;*

(iii) *If $g \sim x_j$ for some $j \geq s+1$, then for any $i \leq s$, $N(x_i) \cap V\left(\overrightarrow{P}(g, z')\right) = \emptyset$.*

Proof of Fact 7. (i) Suppose to the contrary that there exists a vertex z'' in $N(x_j) \cap V\left(\overrightarrow{P}(g, z')\right)$ for some $j \geq i+1$. Then

$$u\overrightarrow{P}gx_i\overleftarrow{P}z''x_j\overleftarrow{P}a_iP_iwP_ja_j\overrightarrow{P}v$$

is a (u, v)-path passing through all vertices of S, a contradiction.

(ii) Suppose to the contrary that there exists a vertex z'' in $N(x_j) \cap V\left(\overrightarrow{P}(g, z')\right)$ for some $i+1 \leq j \leq s$. Then

$$u\overrightarrow{P}x_ig\overleftarrow{P}a_jP_jwP_ia_i\overrightarrow{P}x_jz''\overrightarrow{P}v$$

is a (u, v)-path passing through all vertices of S, a contradiction.

(iii) Suppose to the contrary that there exists a vertex z'' in $N(x_i) \cap V\left(\overrightarrow{P}(g, z')\right)$ for some $i \leq s$. Then

$$u\overrightarrow{P}x_iz''\overrightarrow{P}x_jg\overleftarrow{P}a_iP_iwP_ja_j\overrightarrow{P}v$$

is a (u, v)-path passing through all vertices of S, a contradiction. This proves Fact 7. □

Let $X := \{x_2, \ldots, x_q\}$. For each $P^* \in \mathcal{P}$, let $|N_X(P^*)|$ be the number of vertices in X that is adjacent to a vertex of P^*.

Fact 8. $\sum_{i=2}^{q} d_{V(P^*)}(x_i) \leq |N_X(P^*)| + |P^*| - 1$, *for each $P^* \in \mathcal{P}$.*

Proof of Fact 8. If $P^* = v$, then $|P^*| = 1$ and $\sum_{i=2}^{q} d_{V(P^*)}(x_i) = |N_X(P^*)| + |P^*| - 1$, as desired.

Next assume that $P^* \neq v$. Let $\ell = |P^*|$ and let $z_1, \ldots, z_\ell$ be the vertices of P^* in the order on $\overrightarrow{P}$. If $j \geq 2$ and $N_X(v_j) \neq \emptyset$, let x_{i_j} denote the vertex in $N_X(v_j)$ such that i_j is maximum. Define

$$N'_X(v_j) = \begin{cases} N_X(v_j) \setminus \{x_{i_j}\} & \text{if } j \geq 2 \text{ and } N_X(v_j) \neq \emptyset, \\ N_X(v_j) & \text{if } j = 1 \text{ or } N_X(v_j) = \emptyset. \end{cases}$$

Then by Fact 7, we have that $N'_X(v_a) \cap N'_X(v_b) = \emptyset$ for any distinct integers $a, b \in \{1, \ldots, \ell\}$. Thus,

$$|N_X(P^*)| \geq \left|\bigcup_{j=1}^{\ell} N'_X(v_j)\right| = \sum_{j=1}^{\ell} |N'_X(v_j)|.$$

Therefore,

$$\sum_{i=2}^{q} d_{V(P^*)}(x_i) = |N_X(v_1)| + \sum_{j=2}^{\ell} |N_X(v_j)| \leq |N'_X(v_1)| + \sum_{j=2}^{\ell} \left(|N'_X(v_j)| + 1\right) \leq |N_X(P^*)| + \ell - 1.$$

This completes the proof of Fact 8. □

Fact 9. *For each x_i $(2 \le i \le q)$, x_i is adjacent to at most q member of $\mathcal{P}$. Hence*

$$\sum_{P^* \in \mathcal{P}} |N_X(P^*)| \le q(q-1).$$

Proof of Fact 9. Suppose to the contrary that there exists a vertex x_i such that x_i is adjacent to $P_1^*, \ldots, P_g^*$, where $g \ge q+1$ and $P_1^*, \ldots, P_g^* \in \mathcal{P}$. For each member $P_b^* \in \mathcal{P}$, choose a vertex $y_{P_b^*} \in N(x_i) \cap V(P_b^*)$. We will find a g-fan from x_i to $S \setminus \{x_i\}$, where $g \ge q+1$.

If $P_b^* = v$, let Q_v be the edge $x_i v$. If $P_b^* \ne v$, write $P_b^* = \overrightarrow{P}[z, z']$. Define

$$Q_{P_b^*} = \begin{cases} x_i y_{P_b^*} \overleftarrow{P} z & \text{if } z \in S, \\ x_i y_{P_b^*} \overleftarrow{P} a_j \overleftarrow{P} x_j & \text{if } z = a_j \in A \setminus S \text{ and } j \in \{1, \ldots, q\} \setminus \{i\}, \\ x_i y_{P_b^*} \overleftarrow{P} a_i P_i w & \text{if } z = a_i \in A \setminus S. \end{cases}$$

The family of these internally disjoint $(x_i, S \setminus \{x_i\})$-paths $Q_{P_1^*}, \ldots, Q_{P_g^*}$, is a g-fan from x_i to $S \setminus \{x_i\}$, where $g \ge q+1$, which contradicts the choice of the vertex w. This proves Fact 9. □

We consider $d_G(w) + \sum_{i=2}^{q} d_G(x_i)$. We assert that

$$\sum_{P^* \in \mathcal{P}} (|N_X(P^*)| - 1) \le q(q-2). \tag{4}$$

If fact, for each $P^* \in \mathcal{P}$, we have $|N_X(P^*)| \le |X| = q-1$. If $|\mathcal{P}| \le q$, then by Fact 2(i),

$$\sum_{P^* \in \mathcal{P}} (|N_X(P^*)| - 1) \le |\mathcal{P}|(q-2) \le q(q-2),$$

as desired, If $|\mathcal{P}| > q$, then Fact 9 implies that

$$\sum_{P^* \in \mathcal{P}} (|N_X(P^*)| - 1) = \sum_{P^* \in \mathcal{P}} |N_X(P^*)| - |\mathcal{P}| \le q(q-1) - |\mathcal{P}| < q(q-2).$$

Thus (4) holds.

By Fact 2(ii), we have

$$\begin{aligned} \sum_{i=2}^{q} \sum_{j=2}^{q} \left| N_G(x_i) \cap V\left(\overrightarrow{P}[x_j, a_j)\right)\right| &= \sum_{i=2}^{q} \left| N_G(x_i) \cap V\left(\overrightarrow{P}[x_i, a_i)\right)\right| \\ &\le \sum_{i=2}^{q} \left(\left|\overrightarrow{P}[x_i, a_i)\right| - 1 \right) = \sum_{i=2}^{q} \left|\overrightarrow{P}[x_i, a_i)\right| - q + 1. \end{aligned} \tag{5}$$

Based on Fact 8 and (4), we have

$$\begin{aligned} \sum_{i=2}^{q} \sum_{P^* \in \mathcal{P}} |N_G(x_i) \cap V(P^*)| &= \sum_{P^* \in \mathcal{P}} \sum_{i=2}^{q} |N_G(x_i) \cap V(P^*)| \\ &\le \sum_{P^* \in \mathcal{P}} (|N_X(P^*)| + |P^*| - 1) \\ &\le q(q-2) + \sum_{P^* \in \mathcal{P}} |P^*|. \end{aligned}$$

This, together with (5), implies

$$\begin{aligned} \sum_{i=2}^{q} d_{V(P)}(x_i) &= \sum_{i=2}^{q} |N_G(x_i) \cap V(P)| \\ &= \sum_{i=2}^{q} \left(\sum_{P^* \in \mathcal{P}} |N_G(x_i) \cap V(P^*)| + \sum_{j=2}^{q} \left| N_G(x_i) \cap V\left(\overrightarrow{P}[x_j, a_j)\right)\right| \right) \\ &\le q(q-2) + \sum_{P^* \in \mathcal{P}} |P^*| + \sum_{i=2}^{q} \left|\overrightarrow{P}[x_i, a_i)\right| - q + 1 \\ &= |P| + q^2 - 3q + 1. \end{aligned}$$

Combining this with Fact 5(i)-(ii) and Fact 6(iii) gives

$$\begin{aligned} d_G(w)+\sum_{i=2}^{q} d_G(x_i) &= d_{V(P)}(w)+d_B(w)+d_R(w)+\sum_{i=2}^{q}\Big(d_{V(P)}(x_i)+d_B(x_i)+d_R(x_i)\Big) \\ &\le q+|P|+q^2-3q+1+|R|+|B|-1 \\ &= n+q^2-2q, \end{aligned}$$

which contradicts the assumption that $\sigma_p(S) \ge n+(p-1)^2$ for every integer $k \le p \le \alpha(G)$. This completes the proof of Theorem 1.4. □

3 Application

In this paper, we mainly study the degree-sum condition for the existence of a path passing through prescribed vertices with two endpoints specified. Next we apply Corollary 1.5 to prove Conjecture 1.1. That is,

Theorem 3.1. *Let G be a k-connected graph of order n, where $k \ge 3$. If*

$$\delta(G) > \frac{n+k^2-2k}{k} \quad \textit{and} \quad \delta(G) \ge \alpha(G)+k-2, \tag{6}$$

then G is hamiltonian-connected.

Proof of Theorem 3.1. If $\alpha(G) < k$, then the degree-sum condition in Corollary 1.5 is trivially satisfied, and thus G is hamiltonian-connected, as desired.

Next we assume that $\alpha(G) \ge k$, and let $k \le p \le \alpha(G)$. By (6), $k\delta(G) > n+k^2-2k$, and thus

$$k\delta(G) \ge n+k^2-2k+1.$$

Also, (6) gives $\delta(G) \ge \alpha(G)+k-2 \ge p+k-2$. Since $p-k \ge 0$, it follows that

$$\begin{aligned} p\delta(G) &= (p-k)\delta(G)+k\delta(G) \\ &\ge (p-k)(p+k-2)+n+k^2-2k+1 \\ &= n+(p-1)^2. \end{aligned}$$

Thus $\sigma_p(G) \ge p\delta(G) \ge n+(p-1)^2$ for every $k \le p \le \alpha(G)$. Then by Corollary 1.5, G is hamiltonian-connected, as desired. This completes the proof of Theorem 3.1. □

Remark 2. The condition in (6) can not be weakened. In fact, if we replace (6) with

$$\delta(G) \ge \frac{n+k^2-2k}{k} \quad \text{and} \quad \delta(G) \ge \alpha(G)+k-2, \tag{7}$$

then for $m \ge k-1$, let $H := K_k \vee kK_m$. Then H satisfies (7) and, as shown in Remark 1, is k-connected and not hamiltonian-connected. On the other hand, if we replace (6) with

$$\delta(G) > \frac{n+k^2-2k}{k} \quad \text{and} \quad \delta(G) \ge \alpha(G)+k-3, \tag{8}$$

then for $t \ge k+1$, let $G' = K_t \vee tK_{k-2}$. Clearly, $\alpha(G') = \kappa(G') = t \ge k+1$, and hence G' is k-connected. Moreover, $\delta(G') = t+k-3 = \alpha(G')+k-3$ and $k\delta(G')-(|G'|+k^2-2k) = t-k \ge 1$. Hence, G' satisfies (8). However, G' is not hamiltonian-connected, because its toughness is at most 1.

Declaration

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
Data availability No data was used for the research described in the article.

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